Final Project: Pick a Project

Computer Science - Week 10

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Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Will Mullen	wmullen@udel.edu
Friend (optional)	n/a	n/a

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

I am going to make a program that addresses the suggested Chemical Engineering goal. It will take in a balanced chemical reaction, starting masses, moles, or concentrations in any unit and produce a theoretical yield and limiting reactant.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

Pearson Stoichiometry Calculator - Doesn't calculate molar mass Calculator Online - Filled with ads and not intuitive WebQC Stoichiometry Calculator - Best implementation, but does not allow for two known masses to be inputted (no way to calculate excess)

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

My implementation would be a better version of the stoichiometry calculators currently available. It would be fast, intuitive, and flexible. I want this program to be viable enough for

chemists to use it alongside their experiments. As the experiment progresses, they would input reactants into the program to get a theoretical calculation of their product's mass and thermal energy.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

The program will consume a balanced reaction (a valid string), convert each term into dataclasses to manipulate the molar coefficients and look up the atomic symbols in a library to determine molar masses, then output a variety of floats that represent theoretical yields, mole conversions, mass conversions, etc.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

A balanced reaction (string), known masses (floats), and units (strings).

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

The computer will reformat the inputted string, convert each term into a dataclass, then calculate stoichiometric products.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

The user will be shown all steps of the calculation, the limiting reactant, the theoretical yield, the remaining masses. This will be formatted in a table so that it is easy to read.

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

Reactant 1, Reactant 2, Product 1, (optional) Product 2, all masses involved (reactants, products, excess, theoretical, legitimate)

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

Reactant 1 I need the atomic symbol and atomic mass. I do not need the chemical name, atomic number, reactivity, solubility, etc.

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

1. Drafter implementation 2. Using a library for all chemical values 3. Restructuring the inputted string for it to be compared to the library.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

I plan on using the LLM to provide me with examples and specific structures rather than actually writing out the code. This will force me to develop my own solutions and only use the LLM to implement those solutions. For example, I am not comfortable with Drafter, so I

will attempt to write out a function, probably fail, turn to the AI to help restructure the code so that it is syntactically correct, and then go through multiple iterations (me writing code, AI restructuring, me rewriting that code so that it is more original, AI restructuring that) until I get something that works.

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

I want to create an intuitive website that I can use for my chemistry class. I want it to be easier for me to use than the other websites I have visited.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

I want to build a simple web tool that calculates the theoretical products of a balanced chemical reaction. What are 3 inputs this website should have, at least 3 numbers I should calculate, and what is 1 part of this website you think would make it unique

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

The AI suggested the website should be able to calculate the percentage by mass of elements in a product, the excess reactants, the limiting reactant, and maybe even a reaction visualizer to express mole ratios.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

I dont think I will be able to animate that. what is a simpler alternative? something that makes my website stand out but is easier to implement

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

I know specifically what calculations I should do, and based on the AI's response, I will show the calculations to my outputs.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

ChatGPT, Google, the Discord, and TAs.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

1. *Read the error message and try to understand what I think is going wrong* 2. *Ask the LLM if I am right and tell it to fix it* 3. *Research the error* 4. *Turn to Discord for help*

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I want to build a simple web tool that calculates the theoretical products of a balanced chemical reaction. What are 3 inputs this website should have, at least 3 numbers I should calculate, and what is 1 part of this website you think would make it unique
For a simple web tool that calculates the theoretical products of a balanced chemical reaction, I would focus on making it easy for students to enter a reaction and understand the results.

1. Three inputs for the website

1. Balanced chemical equation — The user enters a balanced reaction, such as
2. 2H2+O2→2H2O
3. 2H
4. 2
5.
6. +O
7. 2
8.
9. →2H
10. 2
11.
12. O.
13. Reactant amounts — The user enters the amount of each reactant available, preferably in grams or moles.
14. Measurement units — The user selects the units for each reactant, such as grams, moles, or liters for gases under specified conditions.

2. At least three numbers to calculate

1. Theoretical yield: The maximum amount of each product that can form, expressed in grams or moles.
2. Limiting reactant: Identify which reactant runs out first and limits the amount of product formed.
3. Excess reactant remaining: Calculate how much of each excess reactant remains after the reaction finishes.
4. Percent composition of products: Calculate the percentage by mass of each element in a product.
5. Mass of products: Calculate the total theoretical mass of products and compare it with the total mass of reactants to demonstrate conservation of mass.

3. One unique feature

Stoichiometric Calculations

Equation:	$2\ H_2(g)$	+	$O_2(g)$	⟶	$2\ H_2O(l)$
Molecules:	2 molecules H_2	+	1 molecule O_2	⟶	2 molecules H_2O
					
Mass (amu):	4.0 amu H_2	+	32.0 amu O_2	⟶	36.0 amu H_2O
Amount (mol):	2 mol H_2	+	1 mol O_2	⟶	2 mol H_2O
Mass (g):	4.0 g H_2	+	32.0 g O_2	⟶	36.0 g H_2O

The coefficients in the balanced equation give the ratio of *moles* of reactants and products

Interactive reaction visualizer
After the user enters a reaction and reactant quantities, the website displays an animated representation of the reaction. It shows reactant molecules being consumed, identifies the limiting reactant, and illustrates how many product molecules can theoretically form.
For example, if the user enters hydrogen and oxygen to produce water, the animation could show hydrogen and oxygen molecules combining into water molecules, with any excess reactant left over.
This would make the tool more engaging than a standard calculator while helping students visualize the connection between balanced equations, mole ratios, and theoretical yield.
My recommendation: Start with the balanced equation, reactant quantities, and units as your three inputs. Make theoretical yield, limiting reactant, and excess reactant remaining your core calculations. Then add the visualizer as your standout feature.
I dont think I will be able to animate that. what is a simpler alternative? something that makes my website stand out but is easier to implement

A much easier alternative would be a step-by-step reaction breakdown that shows users how the website got its answer instead of just giving them a number.

Unique feature: “Show My Work”

After the user enters a balanced equation and reactant amounts, the website displays the calculation process in three simple steps.

Example: Producing water

Balanced equation

2H2+O2→2H2O
2H
2

+O
2

→2H
2

O
Step 1: Convert grams to moles
Explain how the entered reactant masses are converted into moles.
Step 2: Find the limiting reactant
Compare the available moles using the equation's coefficients and identify which reactant runs out first.
Step 3: Calculate theoretical yield
Show the mole ratio calculation and the final theoretical mass of water.

Why this is a good choice

- Easy to implement: You only need text, equations, and basic calculations—no animation or complicated graphics.
- Educational: Users learn how stoichiometry works rather than blindly trusting the answer.
- Interactive: You could add a simple “Show calculation” button to reveal each step.
- Distinctive: Many calculators give answers, but yours could explain the chemistry behind them.

You could even make the feature a toggle: “Quick Answer” or “Teach Me How.” That is a simple addition that gives your website a clear identity.

This might also interest you.

Ads do not influence the answers you get from ChatGPT. Your chats stay private. Learn about ads and personalization

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.